

Closed State Channels Turn Local Language-Model Competence into Long-Horizon Computation

Anonymous authors

Abstract

Reasoning language models can execute one state update yet fail to compose the same update over a history. We force Qwen models through a one-token discrete state channel and delete used history after each call. A two-bit channel hits its exact 50% information limit; direct transition-closure training makes a redundant four-bit channel 93.25% accurate at horizon 16 under five shifted history distributions, versus 48.50% for matched endpoint training. Predicted codes carry a stable three-bit semantic quotient despite surface-token drift.

Question and contract

A history H induces the sufficient current state $S = f(H) \in \{0, \dots, 7\}$; a held-out final rule F gives $Y = g(S, F)$. Native one-pass execution receives (H, F) . Our interface instead emits one token C , deletes H , and gives only (C, F) to a second call. Repeated execution also requires a learned transition $\hat{T}_u$ for each operation u :

$$\underbrace{Y \perp H \mid (C, F)}_{\text{sufficiency}}, \quad \underbrace{q(\hat{T}_u(c)) = T_u(q(c))}_{\text{semantic closure}}, \quad R = \log_2 |\mathcal{C}|.$$

Here q decodes a token to state. Exact canonicalization asks that equal states emit the same token; the weaker and more useful requirement allows a *fiber* $q^{-1}(s)$ of equivalent tokens. This is a finite, causal version of a predictive sufficient statistic [3], with a declared channel rate rather than an unconstrained hidden vector [1, 2].

Method

Diagnosis. On 1,920 balanced modulo-eight programs, frozen Qwen2.5-32B has perfect Read, one-step Update, and supplied-state use, but only 13.44% h2 one-pass Compose. A two-call self-handoff reaches 76.98%; stepwise decimal execution remains perfect through h4.

Training. Rank-16 LoRA on Qwen2.5-7B uses group-disjoint program contexts. Each semantic program supplies two 256-token forwards and two one-token targets: producer $(H \rightarrow C)$ or $(C, u \rightarrow C')$, and consumer $(C, F \rightarrow Y)$. Prompt tokens carry no loss. Endpoint controls use the same programs, forwards, target count, padding, and optimizer, but train $(H \rightarrow C)$ rather than closure. Evaluation composes two-operation calls to h16, discarding each used block.

Controls and interventions. We compare four codes for eight states (2 bits), one opaque code per state (3 bits), and two tokens per state (4 bits). Producer and consumer contexts are disjoint; test contexts are unseen. We measure exact token accuracy, $I(S; C)$, closure after applying q , same-state agreement before/after quotienting, and whether an independently predicted donor code makes the recipient consumer follow the donor state.

Interface / training	h8	h16	decisive check
decimal recursive	100.0	100.0	also 100% at h32
2-bit compressed	50.0	50.0	exact rate limit
3-bit endpoint	35.0	24.5	enough rate, no closure
3-bit closure	55.0	55.0	entry encoder damaged
4-bit endpoint	54.0	52.5	matched control
4-bit closure	97.8	97.1	+43.9, +44.6 points

Table 1: Matched addition programs; accuracy (%). Context-paired 95% intervals for the 4-bit closure gains are [36.5, 51.0] at h8 and [37.6, 51.5] at h16.

Findings and theory

Rate is causal. With uniform S and four deterministic code classes, every token aliases two states, so an arbitrary one-to-one final table permits at most $4/8 = 50\%$ exact recovery. The trained two-bit channel achieves 50% at every horizon while closing perfectly.

Closure, not endpoint fit, produces reuse. Both 4-bit adapters fit short training, but only transition supervision survives recursive use. Across structured, IID, shuffled, cancellation, and repeated histories, closure scores 97.75/95.25/95.00/93.25% at h2/4/8/16; endpoint training scores 100/81.25/63.50/48.50%. At h16, every family remains 90–100%.

The state is a quotient, not a token. In the matched h16 run, only 2.92% of 4-bit outputs have the wrong state, while 42.40% change only the extra bit. Thus exact-token accuracy is 54.69%, but quotient agreement is 95.10%, $I(S; C) = 2.83$ bits, and predicted donor codes cause the right state-dependent answer 97.50% of the time (random controls: 12.3%). Extra rate supplies representational slack; success depends on closure over semantic classes, not surface invariance.

Claim boundary and next tests

The current result shows a learned, rate-limited, causally used computational interface; it does not yet show native internal state formation or broad reasoning. The decisive next tests are: (i) preserve decimal-to-code entry while training closure; (ii) hold out operation pairs and scale state entropy to obtain a rate-reliability curve; (iii) replace addition with register programs and Horn-rule proof states whose next action depends on the compact frontier; and (iv) confirm three seeds and a second 7–8B family. This connects algorithmic length generalization [4] to a measurable interface law: enough relevant bits, a shared decoder, and closure under every legal update.

References

- [1] C. Shannon. A Mathematical Theory of Communication. *BSTJ*, 1948.
- [2] N. Tishby, F. Pereira, W. Bialek. The Information Bottleneck Method. 2000.
- [3] C. Shalizi, J. Crutchfield. Computational Mechanics: Pattern and Prediction, Structure and Simplicity. *JSP*, 2001.
- [4] P. Veličković, C. Blundell. Neural Algorithmic Reasoning. 2021.